Use the following dataset representing 10 students' performance in a course. Features include:

- Study Hours (weekly average hours studied)
- Attendance Percentage (percentage of classes attended)
- Previous Exam Score (out of 100)
- Gender (Male or Female)
- Extracurricular (Yes or No, indicating participation in activities)

Student ID	Study Hours	Attendance Percentage	Previous Exam Score	Gender	Extra-Curricular Activities	Result
1	5	80	65	Male	No	Pass
2	3	60	50	Female	No	Fail
3	6	70	70	Female	Yes	Pass
4	1	50	40	Male	Yes	Fail
5	4	75	60	Male	Yes	Fail
6	8	90	85	Female	Yes	Fail
7	7	85	80	Male	No	Pass
8	9	95	90	Female	Yes	Pass
9	2	80	50	Male	Yes	Fail
10	10	98	95	Male	Yes	Pass

a) Identify which columns denote categorical and numerical features.

	Numerical	Numerical	Numerical	Categorical	Categorical	
Student ID	Study Hours	Attendance Percentage	Previous Exam Score	Gender	Extra-Curricular Activities	Result
1	5	80	65	Male	No	Pass
2	3	60	50	Female	No	Fail
3	6	70	70	Female	Yes	Pass
4	1	50	40	Male	Yes	Fail
5	4	75	60	Male	Yes	Fail
6	8	90	85	Female	Yes	Fail
7	7	85	80	Male	No	Pass
8	9	95	90	Female	Yes	Pass
9	2	80	50	Male	Yes	Fail
10	10	98	95	Male	Yes	Pass

Use the following dataset representing 10 students' performance in a course. Features include:

- Study Hours (weekly average hours studied)
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Student ID	Study Hours	Attendance Percentage	Previous Exam Score	Gender	Extra-Curricular Activities	Result
1	5	80	65	Male	No	Pass
2	3	60	50	Female	No	Fail
3	6	70	70	Female	Yes	Pass
4	1	50	40	Male	Yes	Fail
5	4	75	60	Male	Yes	Fail
6	8	90	85	Female	Yes	Fail
7	7	85	80	Male	No	Pass
8	9	95	90	Female	Yes	Pass
9	2	80	50	Male	Yes	Fail
10	10	98	95	Male	Yes	Pass

a) Identify which columns denote categorical and numerical features.

Categorical: Gender, Extra-Curricular Activities

Numerical: Study Hours, Attendance Percentage, Previous Exam Score

Gender	Result
Male	Pass
Female	Fail
Female	Pass
Male	Fail
Male	Fail
Female	Fail
Male	Pass
Female	Pass
Male	Fail
Male	Pass

b) Calculate the Gini index for "Gender" and "Extra- curricular Activities" feature columns. Show the steps in your calculations.

$$\text{Gini impurity} = 1 - \sum p(i)^2$$

Male {6 samples, 3 Pass, 3 Fail}

$\text{Pass} = \frac{3}{6}$ $\text{Fail} = \frac{3}{6}$

$$\text{Gini impurity} = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2$$

$=1-0.25-0.25$ $=1-0.5$ **$=0.5$**

Gender	Result
Male	Pass
Female	Fail
Female	Pass
Male	Fail
Male	Fail
Female	Fail
Male	Pass
Female	Pass
Male	Fail
Male	Pass

b) Calculate the Gini index for "**Gender**" and "Extra- curricular Activities" feature columns. Show the steps in your calculations.

$$\text{Gini impurity} = 1 - \sum p(i)^2$$

Female {4 samples, 2 Pass, 2 Fail}

$\text{Pass} = \frac{2}{4}$ $\text{Fail} = \frac{2}{4}$

$$\text{Gini impurity} = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2$$

$=1-0.25-0.25$ $=1-0.5$ **$=0.5$**

$$\text{Weighted Gini} = \left(\frac{6}{10}\right)*0.5 + \left(\frac{4}{10}\right)*0.5$$

$= 0.5$

Extra-Curricular Activities	Result
No	Pass
No	Fail
Yes	Pass
Yes	Fail
Yes	Fail
Yes	Fail
No	Pass
Yes	Pass
Yes	Fail
Yes	Pass

b) Calculate the Gini index for "Gender" and "**Extra- curricular Activities**" feature columns. Show the steps in your calculations.

$$\text{Gini impurity} = 1 - \sum p(i)^2$$

No {3 samples, 2 Pass, 1 Fail}

$$\text{Pass} = \frac{2}{3} \qquad \text{Fail} = \frac{1}{3}$$

$$\text{Gini impurity} = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2$$

$$= 0.44$$

Extra-Curricular Activities	Result
No	Pass
No	Fail
Yes	Pass
Yes	Fail
Yes	Fail
Yes	Fail
No	Pass
Yes	Pass
Yes	Fail
Yes	Pass

b) Calculate the Gini index for "Gender" and "**Extra- curricular Activities**" feature columns. Show the steps in your calculations.

$$\text{Gini impurity} = 1 - \sum p(i)^2$$

Yes {7 samples, 3 Pass, 4 Fail}

$\text{Pass} = \frac{3}{7}$ $\text{Fail} = \frac{4}{7}$

$$\text{Gini impurity} = 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2$$

$$= 0.49$$

$$\text{Weighted Gini} = \left(\frac{3}{10}\right)*0.44 + \left(\frac{7}{10}\right)*0.49$$

$$=\mathbf{0.48}$$

c) Based on part (b) result, which feature is most suited to take as the root node in generating the decision tree? Justify your answer.

Weighted Gini [Gender] = 0.5

Weighted Gini [Extra- curricular Activities] = 0.48

Best Splitting → lowest Gini impurity

Extra-Curricular Activities (lower Gini 0.48 < 0.5, better purity after split)

d) The above dataset was split into training and testing sets, with the first 5 rows used for training and the remaining rows for testing. Based on a classifier trained on this split, the predicted values for the test set (Students 6 to 10) are provided.

Student ID	Actual Result	Predicted Result
6	Fail	Pass
7	Pass	Fail
8	Pass	Pass
9	Fail	Fail
10	Pass	Fail

i. Create the confusion matrix.
ii. Calculate the Accuracy of the classifier.
iii. Interpret what each metric Accuracy, Precision, Recall and F1-Score indicate about the classifier's performance in predicting student passing.

Student ID	Actual Result	Predicted Result	
6	Fail	Pass	**FP**
7	Pass	Fail	**FN**
8	Pass	Pass	**TP**
9	Fail	Fail	**TN**
10	Pass	Fail	**FN**

		Predicted	
		Pass	Fail
Actual	Pass	1 [TP]	2 [FN]
	Fail	1 [FP]	1 [TN]

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Accuracy} = \frac{1 + 1}{1 + 1 + 1 + 2}$$

Accuracy = 0.40

		Predicted	
		Pass	Fail
Actual	Pass	1 [TP]	2 [FN]
	Fail	1 [FP]	1 [TN]

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{1}{1+1} = \frac{1}{2} = 0.50$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{1}{1+2} = \frac{1}{3} \approx 0.33$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.50 \times 0.33}{0.50 + 0.33} = 0.40$$

		Predicted	
		Pass	Fail
Actual	Pass	1 [TP]	2 [FN]
	Fail	1 [FP]	1 [TN]

- Accuracy (0.40): % correct predictions.
- Precision (0.50): % predicted Pass that are true.
- Recall (0.33): % true Pass captured.
- F1 (0.40): balance of precision/recall.

- Classifier has low accuracy, moderate precision, low recall. Not a good classifier.

e) Calculate the means and standard deviations for each numerical column you identified in part (a).

Study Hours
5
3
6
1
4
8
7
9
2
10

$$\mu = \frac{5+3+6+1+4+8+7+9+2+10}{10} = \frac{55}{10} = 5.5$$

$$(5-5.5)^2 = 0.25$$
$$(3-5.5)^2 = 6.25$$
$$(6-5.5)^2 = 0.25$$
$$(1-5.5)^2 = 20.25$$
$$(4-5.5)^2 = 2.25$$
$$(8-5.5)^2 = 6.25$$
$$(7-5.5)^2 = 2.25$$
$$(9-5.5)^2 = 12.25$$
$$(2-5.5)^2 = 12.25$$
$$(10-5.5)^2 = 20.25$$

$$\sum (x_i - \bar{x})^2 = 82.5$$

$$\text{SD} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

$$= \sqrt{\frac{82.5}{10}} = \sqrt{8.25}$$

$$= 2.87$$

e) Calculate the means and standard deviations for each numerical column you identified in part (a).

Study Hours	Attendance Percentage	Previous Exam Score
5	80	65
3	60	50
6	70	70
1	50	40
4	75	60
8	90	85
7	85	80
9	95	90
2	80	50
10	98	95

Column	Mean	STD
Study Hours	5.50	2.87
Attendance	78.30	14.39
Prev Score	68.50	17.76

f) Create the scaled dataset using Standardization method only for the numerical columns. ($Z = \frac{x-\mu}{\sigma}$)

Study Hours	$\boldsymbol{x - \mu}$	$\boldsymbol{Z = \frac{x - \mu}{\sigma}}$
5	5 - 5.5 = -0.5	-0.5/2.87= -0.17
3	3 - 5.5 = -2.5	-2.5/2.87= -0.87
6	0.5	0.17
1	-4.5	-1.57
4	-1.5	-0.52
8	2.5	0.87
7	1.5	0.52
9	3.5	1.22
2	-3.5	-1.22
10	4.5	1.57

f) Create the scaled dataset using Standardization method only for the numerical columns. ($Z = \frac{x-\mu}{\sigma}$)

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
2	-0.87	-1.27	-1.04
3	0.17	-0.58	0.08
4	-1.57	-1.97	-1.60
5	-0.52	-0.23	-0.48
6	0.87	0.81	0.93
7	0.52	0.47	0.65
8	1.22	1.16	1.21
9	-1.22	0.12	-1.04
10	1.57	1.37	1.49

g) Compute the Euclidean distances between **Student 1** and **Student 6** and Student 1 and Student 9 using the standardized dataset. Show the steps in your calculations.

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
6	0.87	0.81	0.93

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance: Student 1 ↔ Student 6

$$\begin{aligned} d_{1,6} &= \sqrt{(0.87 - (-0.17))^2 + (0.81 - 0.12)^2 + (0.93 - (-0.20))^2} \\ &= \sqrt{(1.04)^2 + (0.69)^2 + (1.13)^2} \\ &= \sqrt{1.0816 + 0.4761 + 1.2769} \\ &= \sqrt{2.8346} \approx \mathbf{1.68}. \end{aligned}$$

g) Compute the Euclidean distances between Student 1 and Student 6 and **Student 1** and **Student 9** using the standardized dataset. Show the steps in your calculations.

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
9	-1.22	0.12	-1.04

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance: Student 1 ↔ Student 9

$$\begin{aligned} d_{1,9} &= \sqrt{(-1.22 - (-0.17))^2 + (0.12 - 0.12)^2 + (-1.04 - (-0.20))^2} \\ &= \sqrt{(-1.05)^2 + 0^2 + (-0.84)^2} \\ &= \sqrt{1.1025 + 0 + 0.7056} \\ &= \sqrt{1.8081} \approx \mathbf{1.34}. \end{aligned}$$

g) Compute the Euclidean distances between Student 1 and Student 6 and **Student 1** and **Student 9** using the standardized dataset. Show the steps in your calculations.

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
9	-1.22	0.12	-1.04

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance: Student 1 ↔ Student 9

$$\begin{aligned} d_{1,9} &= \sqrt{(-1.22 - (-0.17))^2 + (0.12 - 0.12)^2 + (-1.04 - (-0.20))^2} \\ &= \sqrt{(-1.05)^2 + 0^2 + (-0.84)^2} \\ &= \sqrt{1.1025 + 0 + 0.7056} \\ &= \sqrt{1.8081} \approx \mathbf{1.34}. \end{aligned}$$

h) Based on the results from part (g), identify which student is the most suitable to merge with Student 1 into the same cluster. Justify your answer.

Distance [Student 1 ↔ Student 6] : 1.68

Distance [Student 1 ↔ Student 9] : 1.34

Student 9 (smaller distance 1.34 < 1.68) Smaller distance implies clusters are closer to one another

Student 9 is closest to student 1

We can merge them together

i) For k-means clustering (k=2) on the standardized dataset obtained in part(f), the clusters and their centroids are given below.

Cluster	Student ID	Centroid Coordinates
Cluster 1 – C1	1, 2, 3, 4, 5	(-0.871, -0.785, -0.867)
Cluster 2 – C2	6, 7, 8, 9, 10	(0.871, 0.785, 0.867)

Answer the following:

i. Compute the inertia (within-cluster sum of squares) for the two clusters separately using the equation given below. Show calculations for each cluster and sum them to get the total inertia.

$$\boldsymbol{Inertia} = \sum_{\boldsymbol{points\ in\ cluster}} (\boldsymbol{Euclidean\ distance\ to\ centroid})^2$$

ii. Compute the Dunn index for the above clustering. Show all calculations.

iii. Explain what the calculated Dunn index value indicates about the quality of the clusters.

Cluster 1 (C1): Students 1, 2, 3, 4, 5

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20

Cluster 1 Centroid = (−0.871, −0.785, −0.867)

ID	Δx = (x − x_c)	Δy = (y − y_c)	Δz = (z − z_c)	(d^2= Δx^2+Δx^2+Δx^2)	Euclidean Distance to Centroid
1	(-0.17 - (-0.871)) = = 0.701	(0.12 - (-0.785)) = = 0.905	(-0.20 - (-0.867)) = 0.667	0.701^2 + 0.905^2 + 0.667^2 **= 1.75**	1.32

Cluster 1 (C1): Students 1, 2, 3, 4, 5

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
1	-0.17	0.12	-0.20
2	-0.87	-1.27	-1.04

Cluster 1 Centroid = (−0.871, −0.785, −0.867)

ID	Δx = (x − x_c)	Δy = (y − y_c)	Δz = (z − z_c)	(d^2= Δx^2+Δx^2+Δx^2)	Euclidean Distance to Centroid
1	(-0.17 - (-0.871)) = = 0.701	(0.12 - (-0.785)) = = 0.905	(-0.20 - (-0.867)) = 0.667	0.701^2 + 0.905^2 + 0.667^2 = 1.75	1.32
2	(-0.87-(-0.871)) = 0.001	(-1.27-(-0.785)= -0.485	(-1.04-(-0.867))= -0.173	0.001^2 + $(-0.485)^2$ + $(-0.173)^2$ =0.27	0.51

Cluster 1 (C1): Students 1, 2, 3, 4, 5

Cluster 1 Centroid = (−0.871, −0.785, −0.867)

ID	Δx = (x − x_c)	Δy = (y − y_c)	Δz = (z − z_c)	(d^2= Δx^2+Δx^2+Δx^2)	Euclidean Distance to Centroid
1	(-0.17 - (-0.871)) = 0.701	(0.12 - (-0.785)) = 0.905	(-0.20 - (-0.867)) = 0.667	$0.701^2 + 0.905^2 + 0.667^2$ = 1.75	1.32
2	(-0.87-(-0.871)) = 0.001	(-1.27-(-0.785)= -0.485	(-1.04-(-0.867))= -0.173	$0.001^2 + (-0.485)^2 + (-0.173)^2$ =0.27	0.51
3	(0.17-(-0.871))= 1.041	(-0.58-(-0.785))= 0.205	(0.08-(-0.867)= 0.947	2.02	1.42
4	(-1.57-(-0.871)) = -0.699	(-1.97-(-0.785))= -1.185	(-1.60-(-0.867))= -0.733	2.43	**1.56**
5	((-0.52-(-0.871))= 0.351	(-0.23-(-0.785)= 0.555	(-0.48-(-0.867)= 0.387	0.58	0.76

Inertia(C1)= 1.75+0.27+2.02+2.43+0.58
= 7.05

Cluster 2 (C2): Students 6, 7, 8, 9, 10

Student ID	Study Hours	Attendance Percentage	Previous Exam Score
6	0.87	0.81	0.93

Cluster 2 Centroid = (0.871,0.785,0.867)

ID	Δx = (x − x_c)	Δy = (y − y_c)	Δz = (z − z_c)	(d^2= Δx^2+Δx^2+Δx^2)	Euclidean Distance to Centroid
6	(0.87 − 0.871) = - 0.001	(0.81 − 0.785) = 0.025	(0.93 − 0.867) = 0.063	$-0.001^2 + 0.025^2 + 0.063^2$ = **0.005**	0.07

Cluster 2 (C2): Students 6, 7, 8, 9, 10

Cluster 2 Centroid = (0.871,0.785,0.867)

ID	**Δx = (x − x_c)**	**Δy = (y − y_c)**	**Δz = (z − z_c)**	**(d^2= Δx^2+Δx^2+Δx^2)**	**Euclidean Distance to Centroid**
6	(0.87 − 0.871) = - 0.001	(0.81 − 0.785) = 0.025	(0.93 − 0.867) = 0.063	$-0.001^2 + 0.025^2 + 0.063^2$ = 0.005	0.07
7	-0.351	-0.315	-0.217	0.27	0.52
8	0.349	0.375	0.343	0.38	0.62
9	-2.091	-0.665	-1.907	8.45	**2.91**
10	0.699	0.585	0.623	1.22	1.10

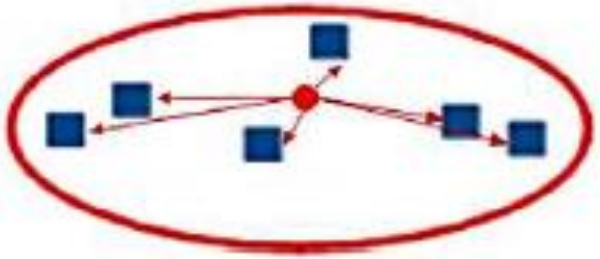
Inertia(C2)=0.005+0.27+0.38+8.45+1.22
=10.33

Total Inertia = Inertia(C1) + Inertia(C2)

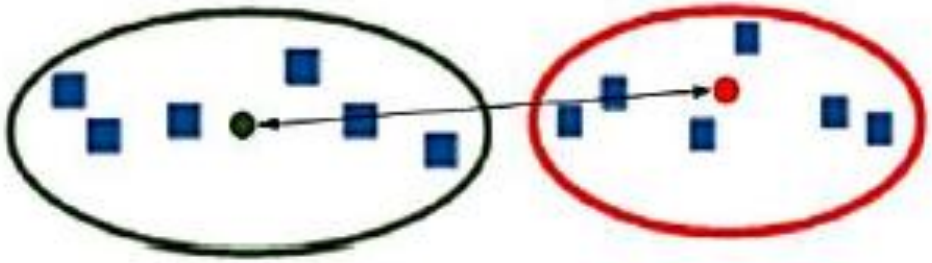
= 7.05+10.33

= 17.38

Compute the Dunn index for the above clustering. Show all calculations.



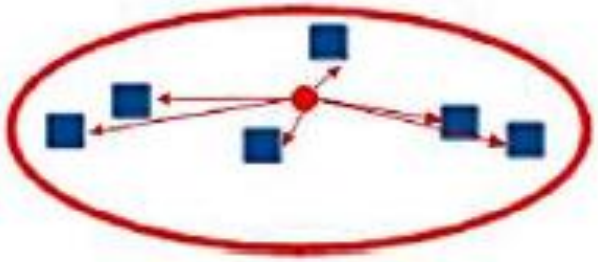
Intra cluster distance



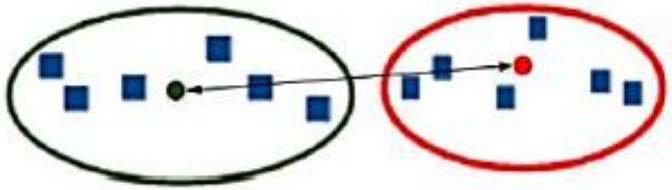
Inter cluster distance

$$\text{Dunn Index} = \frac{\min(\text{Inter cluster distance})}{\max(\text{Intra cluster distance})}$$

Compute the Dunn index for the above clustering. Show all calculations.



Intra cluster distance



Inter cluster distance

Max intra Cluster 1 distance: 1.56

Max intra Cluster 2 distance: 2.91

Max intra-cluster distance overall: **2.91**

C1=(−0.871,−0.785,−0.867)

C2=(0.871,0.785,0.867)

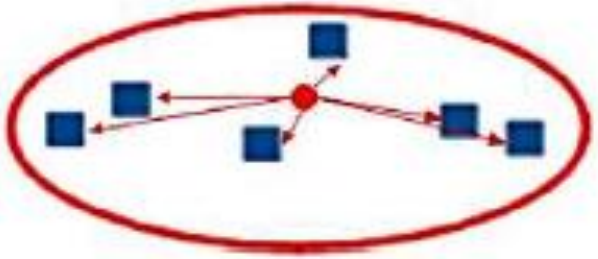
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$d = \sqrt{(0.871 - (-0.871))^2 + (0.785 - (-0.785))^2 + (0.867 - (-0.867))^2}$$

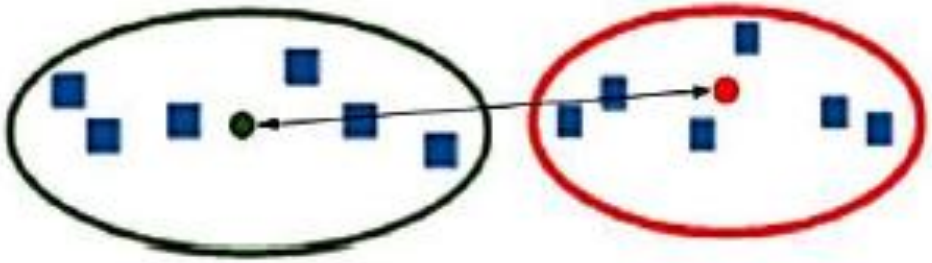
$$d = \sqrt{(1.742)^2 + (1.570)^2 + (1.734)^2} \quad = \sqrt{8.509}$$

Min inter-cluster distance = **2.92**

Compute the Dunn index for the above clustering. Show all calculations.



Intra cluster distance



Inter cluster distance

$$\text{Dunn Index} = \frac{\min(\text{Inter cluster distance})}{\max(\text{Intra cluster distance})}$$

Dunn Index = 2.92 / 2.91 ≈ 1.00

i. Explain what the calculated Dunn index value indicates about the quality of the clusters.

The Dunn Index of 1.00 indicates moderate cluster quality.

The clusters are reasonably separated (inter-cluster distance 2.92 ≈ maximum intra-cluster distance 2.91), but not highly compact, suggesting some overlap or spread, particularly in Cluster 2.

The clustering is acceptable but could be improved for better separation and compactness.

Imagine you are a junior data analyst at a small online bookstore that sells fiction, non-fiction, and autobiography. A sample raw dataset of 5 customers and their corresponding standardized datasets are shown below.

Customer	Age (raw)	Books/Year (raw)	Spending/Book (raw)
C1	20	6	$14.50
C2	30	14	$25.75
C3	40	8	$17.25
C4	50	6	$14.50
C5	30	14	$28.00

Standardization stats:

- Age
  - **mean** = 34 **std** = 10.20
- Books
  - **mean** = 10 **std** = 3.67
- Spending
  - **mean** = $20 **std** = 5.75

Dataset has three features: age (in years), number of books bought per year, and average spending per book (in dollars). There are no pre-labeled groups, but the goal is to group customers to suggest better book recommendations. For example, young frequent buyers might get teen fiction suggestions.

You'll use agglomerative clustering to find initial groups and then K-Nearest Neighbors to assign new customers to those groups.

Apply agglomerative clustering with **complete linkage** and Euclidean distance to group them into clusters. Compute distance matrices and iteratively merge closest clusters using complete linkage until one cluster is formed.

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09				
C3					
C4					
C5					

Distance between C1 and C2

$$d(\mathrm{C1}, C2) = \sqrt{\left(-1.37-(-0.39)\right)^2 + \left(-1.09-1.09\right)^2 + \left(-0.96-1.00\right)^2}$$

$$= \sqrt{\left(-0.98\right)^2 + \left(-2.18\right)^2 + \left(-1.96\right)^2}$$

$$= \sqrt{0.9604 + 4.7524 + 3.8416}$$

$$= \sqrt{9.5544} = 3.09$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09				
C3	2.09				
C4					
C5					

Distance between C1 and C3

$$d(C1, C3) = \sqrt{\left(-1.37 - 0.59\right)^2 + \left(-1.09 - (-0.54)\right)^2 + \left(-0.96 - (-0.48)\right)^2}$$

$$= \sqrt{\left(-1.96\right)^2 + \left(-0.55\right)^2 + \left(-0.48\right)^2}$$

$$= \sqrt{3.8416 + 0.3025 + 0.2304}$$

$$= \sqrt{4.3745} = \boxed{2.09}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09				
C3	2.09				
C4	2.94				
C5					

Distance between C1 and C4

$$d(C1, C4) = \sqrt{\left(-1.37 - 1.57\right)^2 + \left(-1.09 - (-1.09)\right)^2 + \left(-0.96 - (-0.96)\right)^2}$$

$$= \sqrt{\left(-2.94\right)^2 + \left(0\right)^2 + \left(0\right)^2}$$

$$= \sqrt{8.6436} = \boxed{2.94}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09				
C3	2.09				
C4	2.94				
C5	3.35				

Distance between C1 and C5

$$d(C1, C5) = \sqrt{\left(-1.37 - (-0.39)\right)^2 + \left(-1.09 - 1.09\right)^2 + \left(-0.96 - 1.39\right)^2}$$

$$= \sqrt{\left(-0.98\right)^2 + \left(-2.18\right)^2 + \left(-2.35\right)^2}$$

$$= \sqrt{0.9604 + 4.7524 + 5.5225}$$

$$= \sqrt{11.2353} = \boxed{3.35}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41			
C4	2.94				
C5	3.35				

Distance between C2 and C3

$$d(C,2C3) = \sqrt{\left(-0.39 - 0.59\right)^2 + \left(1.09 - (-0.54)\right)^2 + \left(1.00 - (-0.48)\right)^2}$$

$$= \sqrt{\left(-0.98\right)^2 + \left(1.63\right)^2 + \left(1.48\right)^2}$$

$$= \sqrt{0.9604 + 2.6569 + 2.1904}$$

$$= \sqrt{5.8077} = \boxed{2.41}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41			
C4	2.94	3.53			
C5	3.35				

Distance between C2 and C4

$$d(C,2C4) = \sqrt{\left(-0.39-1.57\right)^2+\left(1.09-(-1.09)\right)^2+\left(1.00-(-0.96)\right)^2}$$

$$= \sqrt{\left(-1.96\right)^2+\left(2.18\right)^2+\left(1.96\right)^2}$$

$$= \sqrt{3.8416+4.7524+3.8416}$$

$$= \sqrt{12.4356} = \boxed{3.53}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41			
C4	2.94	3.53			
C5	3.35	0.39			

Distance between C2 and C5

$$
\begin{aligned}
d(C2, C5) &= \sqrt{(-0.39 - (-0.39))^2 + (1.09 - 1.09)^2 + (1.00 - 1.39)^2} \\
&= \sqrt{0^2 + 0^2 + (-0.39)^2} \\
&= \sqrt{0 + 0 + 0.1521} \\
&= \sqrt{0.1521} = \boxed{0.39}
\end{aligned}
$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22		
C5	3.35	0.39			

Distance between C3 and C4

$$d(C,3C4) = \sqrt{\left(0.59 - 1.57\right)^2 + \left(-0.54 - (-1.09)\right)^2 + \left(-0.48 - (-0.96)\right)^2}$$

$$= \sqrt{\left(-0.98\right)^2 + \left(0.55\right)^2 + \left(0.48\right)^2}$$

$$= \sqrt{0.9604 + 0.3025 + 0.2304}$$

$$= \sqrt{1.4933} = \boxed{1.22}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67		

Distance between C3 and C5

$$d(C,3C5) = \sqrt{\left(0.59-(-0.39)\right)^2+\left(-0.54-1.09\right)^2+\left(-0.48-1.39\right)^2}$$

$$= \sqrt{\left(0.98\right)^2+\left(-1.63\right)^2+\left(-1.87\right)^2}$$

$$= \sqrt{0.9604+2.6569+3.4969}$$

$$= \sqrt{7.1142} = \boxed{2.67}$$

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67	3.76	0

Distance between C4 and C5

$$d(C,4C5) = \sqrt{\left(1.57-(-0.39)\right)^2+\left(-1.09-1.09\right)^2+\left(-0.96-1.39\right)^2}$$

$$= \sqrt{\left(1.96\right)^2+\left(-2.18\right)^2+\left(-2.35\right)^2}$$

$$= \sqrt{3.8416+4.7524+5.5225}$$

$$= \sqrt{14.1165} = \boxed{3.76}$$

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67	3.76	0

Merge: C2 and C5 at distance 0.39, forming {C2, C5}

	C1	C2, C5	C3	C4
C1	0			
C2, C5	?	0		
C3	2.09	?	0	
C4	2.94	?	1.22	0

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67	3.76	0

Distances from the new cluster to others are the maximum of the corresponding pairs (complete linkage):

$$d(C1,\{C2C5\}) = \max\{ d(C1,C2)=3.09,\ d(C1,C5)=3.35 \} = \mathbf{3.35}$$

$$d(C3,\{C2C5\}) = \max\{ d(C3,C2)=2.41,\ d(C3,C5)=2.67 \} = \mathbf{2.67}$$

$$d(C4,\{C2C5\}) = \max\{ d(C4,C2)=3.53,\ d(C4,C5)=3.76 \} = \mathbf{3.76}$$

	C1	C2, C5	C3	C4
C1	0			
C2, C5	3.35	0		
C3	2.09	2.67	0	
C4	2.94	3.76	1.22	0

	C1	C2, C5	C3	C4
C1	0			
C2, C5	3.35	0		
C3	2.09	2.67	0	
C4	2.94	3.76	1.22	0

Merge: C3 and C4 at distance 1.22, forming {C3, C4}.

	C1	C2, C5	C3, C4
C1	0		
C2, C5	3.35	0	
C3, C4	?	?	0

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67	3.76	0

Distances from the new cluster to others are the maximum of the corresponding pairs (complete linkage):

$$d(C1,\{C3C4\}) = \max\{ d(C1,C3)=2.09,\ d(C1,C4)=2.94 \} = \mathbf{2.94}$$

$$d(\{C2C5\},\{C3C4\}) = \max\{d(C2,C3)=2.41,\ d(C2,C4)=3.53,\ d(C5,C3)=2.67,\ d(C5,C4)=3.76 \} = \mathbf{3.76}$$

	C1	C2, C5	C3, C4
C1	0		
C2, C5	3.35	0	
C3, C4	2.94	3.76	0

	C1	C2, C5	C3, C4
C1	0		
C2, C5	3.35	0	
C3, C4	2.94	3.76	0

Merge: C1 and {C3, C4} at distance 2.94, forming {C1, C3, C4}

	C1, C3, C4	C2, C5
C1, C3, C4	0	
C2, C5	?	0

	C1	C2	C3	C4	C5
C1	0				
C2	3.09	0			
C3	2.09	2.41	0		
C4	2.94	3.53	1.22	0	
C5	3.35	0.39	2.67	3.76	0

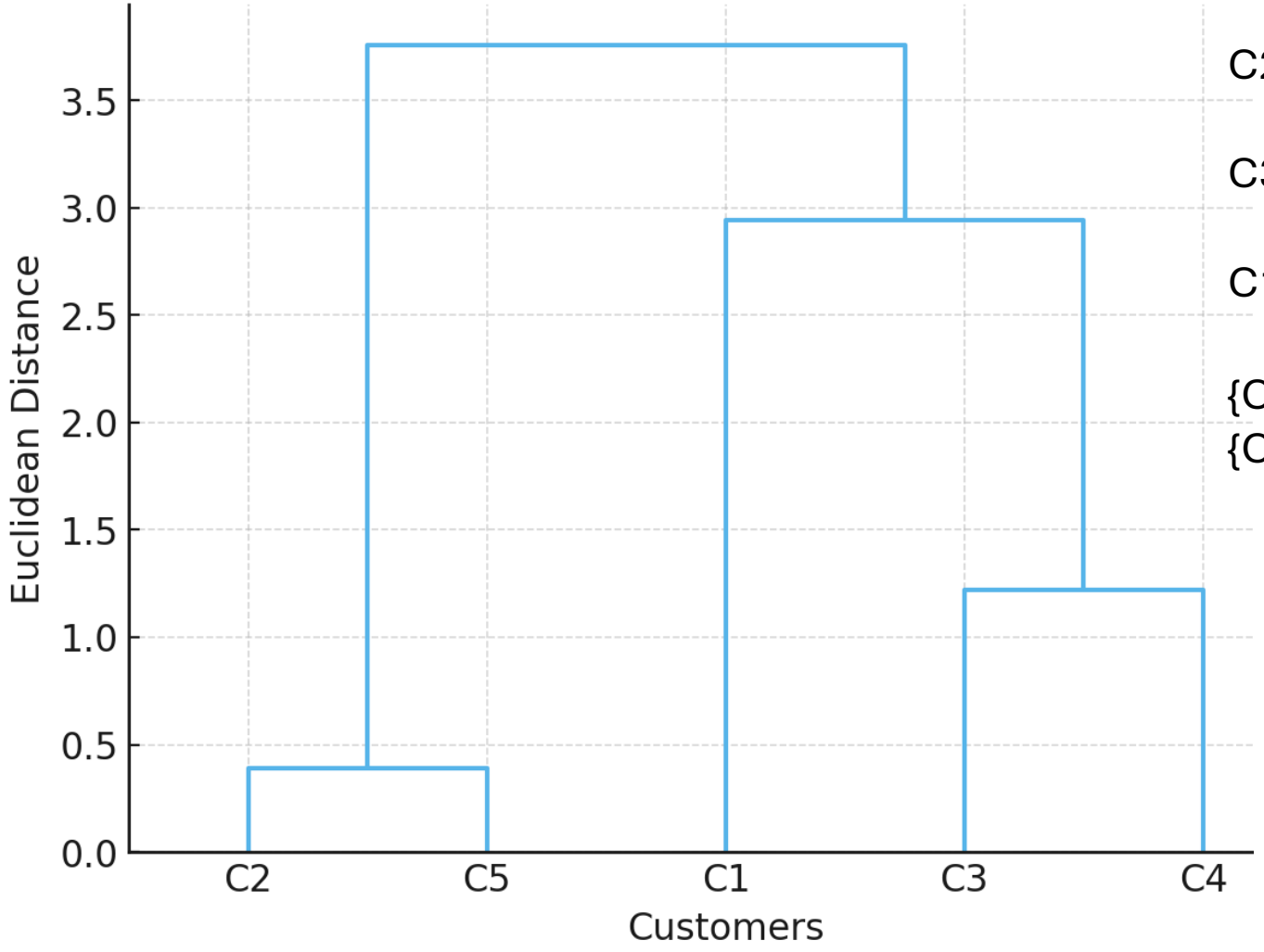
Distances from the new cluster to others are the **minimum** of the corresponding pairs (single linkage):

$$d(\{\text{C1C3C4}\},\{\text{C2C5}\}) = \max\{\underbrace{d(\text{C1},\text{C2})}_{3.09},\underbrace{d(\text{C1},\text{C5})}_{3.35},\underbrace{d(\text{C3},\text{C2})}_{2.41},\underbrace{d(\text{C3},\text{C5})}_{2.67},\underbrace{d(\text{C4},\text{C2})}_{3.53},\underbrace{d(\text{C4},\text{C5})}_{\mathbf{3.76}}\} = \mathbf{3.76}$$

	C1, C3, C4	C2, C5
C1, C3, C4	0	
C2, C5	3.76	0

Merge: {C1, C3, C4} and {C2, C5} at distance 3.76, forming {C1, C2, C3, C4, C5}

a) Based on part (a), sketch the dendrogram. Represent the horizontal line to identify 3 clusters and justify your answer.

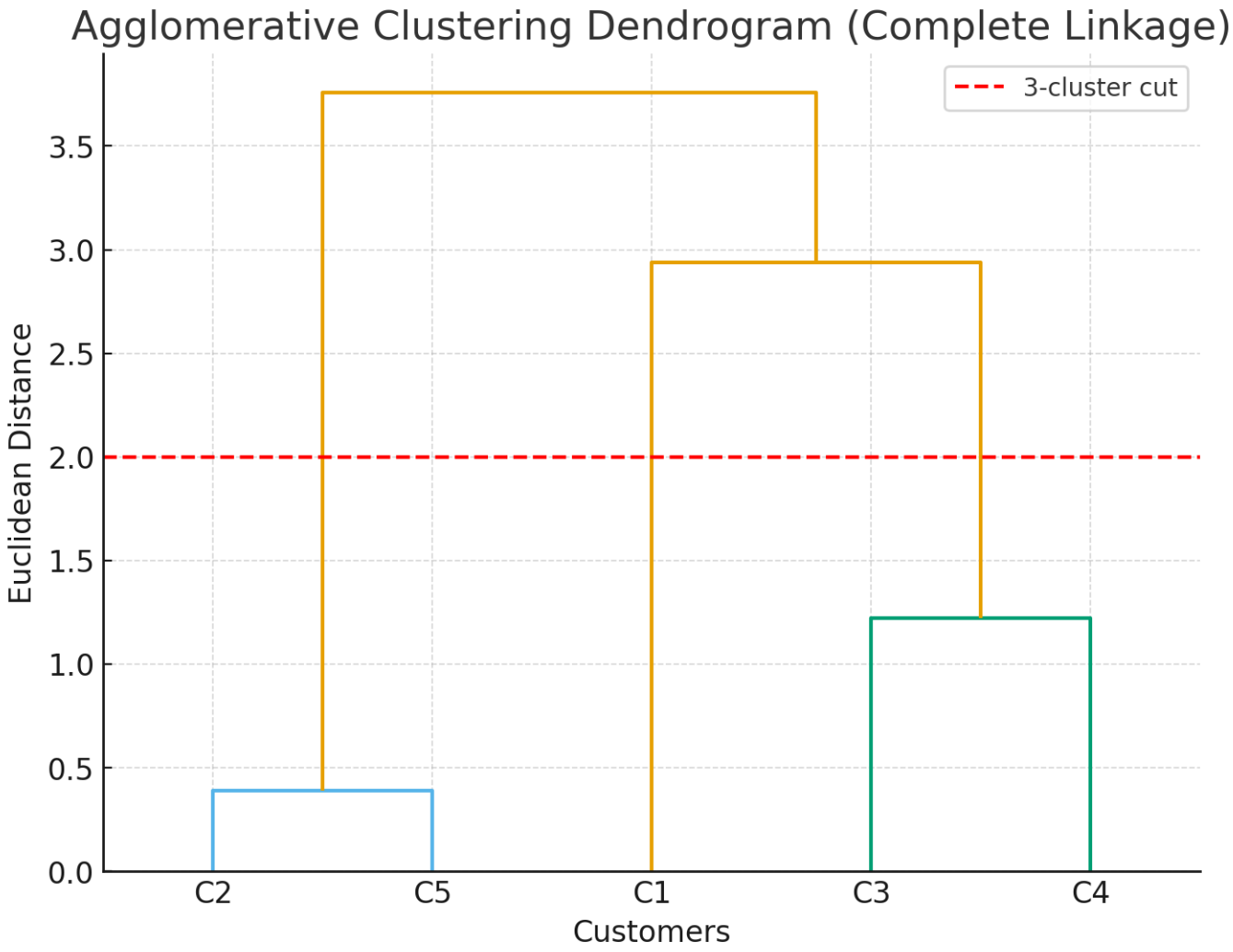


C2 and C5 at distance 0.39, forming {C2, C5}

C3 and C4 at distance 1.22, forming {C3, C4}

C1 and {C3, C4} at distance 2.94, forming {C1, C3, C4}

{C1, C3, C4} and {C2, C5} at distance 3.76, forming {C1, C2, C3, C4, C5}



To form 3 clusters, cut the dendrogram at a height between 1.22 and 2.94 (e.g., at ~2.0):

Cluster 1: {C2, C5}

Cluster 2: {C3, C4}

Cluster 3: {C1}

Using the clusters identified from part (c) as labels, describe how to apply KNN (with K=3 and Euclidean distance) to classify a new customer: **30-year-old who buys 14 books a year at $26 each**.

Given standardization statistics

- Age mean = 34, std = 10.20
- Books mean = 10, std = 3.67
- Spending mean = $20, std = 5.75

Compute $z = \frac{x - mean}{std}$:

$$z_{Age} = \frac{30 - 34}{10.20} = -0.39,$$

$$z_{Books} = \frac{14 - 10}{3.67} = 1.09,$$

$$z_{Spend} = \frac{26 - 20}{5.75} = 1.04.$$

New customer (standardized): **(-0.39, 1.09, 1.04)**

New customer (standardized): **(-0.39, 1.09, 1.04)**

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

To C1 (−1.37, −1.09, −0.96)

$$\sqrt{\left(-0.39-(-1.37)\right)^2+\left(1.09-(-1.09)\right)^2+\left(1.04-(-0.96)\right)^2}$$

$$=\sqrt{\left(0.98\right)^2+\left(2.18\right)^2+\left(2.00\right)^2}=\sqrt{0.96+4.75+4.00}$$

$$=\sqrt{9.71}\approx\boxed{3.12}$$

To C2 (−0.39, 1.09, 1.00)

$$\sqrt{\left(-0.39-(-0.39)\right)^2+\left(1.09-1.09\right)^2+\left(1.04-1.00\right)^2}$$

$$=\sqrt{\left(0\right)^2+\left(0\right)^2+\left(0.04\right)^2}=\sqrt{0.0016}\approx\boxed{0.04}$$

New customer (standardized): **(-0.39, 1.09, 1.04)**

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

To C3 (0.59, −0.54, −0.48)

$$\sqrt{\left(-0.39-0.59\right)^2+\left(1.09-(-0.54)\right)^2+\left(1.04-(-0.48)\right)^2}$$

$$=\sqrt{\left(-0.98\right)^2+\left(1.63\right)^2+\left(1.52\right)^2}=\sqrt{0.96+2.66+2.31}$$

$$=\sqrt{5.93}\approx\boxed{2.44}$$

To C4 (1.57, −1.09, −0.96)

$$\sqrt{\left(-0.39-1.57\right)^2+\left(1.09-(-1.09)\right)^2+\left(1.04-(-0.96)\right)^2}$$

$$=\sqrt{\left(-1.96\right)^2+\left(2.18\right)^2+\left(2.00\right)^2}=\sqrt{3.84+4.75+4.00}$$

$$=\sqrt{12.59}\approx\boxed{3.55}$$

New customer (standardized): **(-0.39, 1.09, 1.04)**

Customer	Age	Books/Year	Spending/Book
C1	-1.37	-1.09	-0.96
C2	-0.39	1.09	1.00
C3	0.59	-0.54	-0.48
C4	1.57	-1.09	-0.96
C5	-0.39	1.09	1.39

To C5 (−0.39, 1.09, 1.39)

$$\sqrt{\left(-0.39-(-0.39)\right)^2+\left(1.09-1.09\right)^2+\left(1.04-1.39\right)^2}$$

$$=\sqrt{\left(0\right)^2+\left(0\right)^2+\left(-0.35\right)^2}=\sqrt{0.1225}\approx\boxed{0.35}$$

Cluster 1: {C2, C5}
Cluster 2: {C3, C4}
Cluster 3: {C1}

Customer	Distance to New	Cluster
C1	3.12	3
C2	0.04	1
C3	2.44	2
C4	3.55	2
C5	0.35	1

Select 3 nearest neighbors:

C2: 0.04 (Cluster 1)

C5: 0.35 (Cluster 1)

C3: 2.44 (Cluster 2)

Majority vote: 2 neighbors (C2, C5) from Cluster 1, 1 (C3) from Cluster 2

The new customer is classified into Cluster 1